

A Self-Contained Proof of Graffiti.pc Conjecture 84:

$$t(G) \geq 2r(G)/\delta(G)$$

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Abstract

For a finite connected graph G , let $t(G)$ denote the largest order of an induced subtree, let $r(G)$ denote the radius, and let $\delta(G)$ denote the minimum degree. Graffiti.pc Conjecture 84 (West's catalogue, 2004) asserts

$$t(G) \geq \frac{2r(G)}{\delta(G)}.$$

We prove the equivalent integral inequality $2r(G) \leq t(G)\delta(G)$, via the leaf lemma: every finite connected graph with a leaf contains an induced tree on at least $2r(G)$ vertices. The presentation is fully self-contained. In particular the Erdős-Saks-Sós induced-path theorem is not used (a geodesic bound suffices for the case $\delta \geq 2$), and the radius-drop criterion together with the structure theorem for vertex-radius-decreasing graphs $G = S_\ell(B)$ —classically due to Gliviak and Fajtlowicz—are re-proved here from elementary principles.

1 The conjecture and overview

All graphs are finite, simple, and undirected. For a connected graph G write

$$\text{ecc}_G(v) = \max_{x \in V(G)} d_G(v, x), \quad r(G) = \min_{v \in V(G)} \text{ecc}_G(v),$$

and let $\delta(G)$ be the minimum degree. Let $t(G)$ be the maximum cardinality of a vertex set inducing a tree. A *leaf* is a vertex of degree one, $p(u)$ denotes the neighbour of a leaf u , and v is a *cut vertex* if $G - v$ is disconnected; otherwise v is a non-cut vertex.

Fact 1.1 (Graffiti.pc #84; West [7]). If G is connected, then $t(G) \geq 2r(G)/\delta(G)$.

For connected G with $|V(G)| \geq 2$ we have $\delta(G) \geq 1$, so the statement is equivalent to the integral inequality

$$2r(G) \leq t(G)\delta(G). \tag{1}$$

(The one-vertex graph makes $\delta = 0$; (1) then reads $0 \leq 0$ and is trivial, so we always assume $|V(G)| \geq 2$.)

The bound is tight: the path P_{2r} has $\delta = 1$ and $t = 2r$. Hence the case $\delta(G) = 1$ cannot be improved, and it is the whole difficulty:

Lemma 1.2 (Leaf lemma). *Every finite connected graph with a leaf contains an induced tree on at least $2r(G)$ vertices.*

Indeed, if $\delta(G) \geq 2$ then $t(G) \geq r(G)+1$ by Lemma 2.1 below, whence $t(G)\delta(G) \geq 2(r+1) > 2r$; if $\delta(G) = 1$ the leaf lemma gives $t(G)\delta(G) = t(G) \geq 2r$. This proves Conjecture 1.1 modulo Lemma 1.2, which is proved in Section 5.

The proof of the leaf lemma has two branches, according to whether deleting the leaf lowers the radius. One branch is governed by the following classical notion.

Definition 1.3. A connected nontrivial graph G is *vertex-radius-decreasing* (vrd) if $r(G - v) < r(G)$ for every non-cut vertex v of G . By Lemma 2.2 the inequality is automatically an equality: $r(G - v) = r(G) - 1$.

Definition 1.4. For a connected graph B and $\ell \geq 1$, let $S_\ell(B)$ be the graph obtained from B by attaching to *every* vertex of B a pendant path of length ℓ (a path on ℓ new vertices, whose last vertex is a leaf of $S_\ell(B)$), using that vertex as the first endpoint.

Theorem 1.5 (Structure of vrd graphs with a cut vertex). *If G is vrd, has a cut vertex, and $r(G) \geq 2$, then $G = S_\ell(B)$ for some $\ell \geq 1$ and some connected block B (of order at least two), every vertex of which has degree $\ell + 1$ in G ; moreover*

$$r(G) = r(B) + \ell.$$

Consequently $t(G) \geq 2r(G)$ (Corollary 4.9).

Classically this is due to Glivak [4] (see also the survey [5], Theorem 4, and Swart [6, Theorem 2.2.10]), with an ingredients credited to Fajtlowicz [3]. Here we prove Theorem 1.5 from scratch (Section 4). The non-cut-vertex argument below uses paths directly, without a block tree. The proof of the leaf lemma also uses a *rooted* form of Chung's induced-path construction (Lemma 3.1), proved in full in Section 3, including two details that are usually left implicit.

Dependency account

- Section 2: elementary facts, all proved.
- Section 3: the rooted Chung lemma, self-contained.
- Section 4: Theorem 1.5 and its corollary, self-contained.
- Section 5: the leaf lemma, from the two preceding items.
- Section 6: Conjecture 1.1.

No result of [1, 3, 4, 5, 6] is used in any proof; those works are cited only as historical credits.

2 Elementary facts

Throughout, G is a finite connected graph with n vertices; v is a vertex with $G - v$ connected and nonempty when a radius $r(G - v)$ is mentioned.

Lemma 2.1 (Geodesics; the bound $t \geq r + 1$). *Every geodesic path induces a path subgraph (no chord joins two of its vertices non-consecutively). Consequently $\text{rad}(G) \leq \text{diam}(G)$ and $t(G) \geq \text{diam}(G) + 1 \geq r(G) + 1$.*

Proof. A chord would shorten the geodesic. Vertices at distance $\text{diam}(G)$ are joined by a geodesic, which is induced, of order $\text{diam}(G) + 1$; and $\text{rad}(G) \leq \text{ecc}(x) \leq \text{diam}(G)$ for every x . $\square$

Lemma 2.2 (Radius drops by at most one). *If $G - v$ is connected and nonempty then $r(G) \leq r(G - v) + 1$.*

Proof. Let c' be a centre of $G - v$ and $\rho = r(G - v)$. For $x \neq v$, $d_G(c', x) \leq d_{G-v}(c', x) \leq \rho$. The vertex v has a neighbour w in $G - v$, so $d_G(c', v) \leq 1 + \rho$. Hence $\text{ecc}_G(c') \leq \rho + 1$. $\square$

Lemma 2.3 (Leaves). *Let z be a leaf of G with $|V(G)| \geq 3$.*

- (i) *Every path with both ends $\neq z$ avoids z ; hence distances between vertices of $G - z$ agree in G and $G - z$.*
- (ii) *$r(G - z) \leq r(G)$: a centre c of G is not a leaf, and $\text{ecc}_{G-z}(c) = \text{ecc}_G(c) = r(G)$ by (i).*
- (iii) *z is not central: $\text{ecc}(z) = 1 + \text{ecc}(p(z)) \geq 1 + r(G) > r(G)$.*
- (iv) *$p(z)$ is a cut vertex (for $|V(G)| \geq 3$).*

Proof. (i) A path through z must enter and leave z along distinct edges, but z has degree one. (ii) $z \neq c$ by (iii). (iii) Clear. (iv) z is isolated in $G - p(z)$, while $G - p(z)$ has other vertices. $\square$

Lemma 2.4 (Unique eccentric points, I: the drop). *Call x the unique eccentric point (UEP) of c if $d(c, x) = \text{ecc}(c)$ and $d(c, y) < \text{ecc}(c)$ for all $y \neq x$. If c is central and has UEP v , then $G - v$ is connected and $r(G - v) = r(G) - 1$.*

Proof. Write $r = r(G)$. For $y \neq v$, a c - y path through v has length at least $d(c, v) + 1 \geq r + 1 > r - 1 \geq d(c, y)$; so every c - y geodesic avoids v , and $d_{G-v}(c, y) = d_G(c, y) \leq r - 1$. Thus every vertex of $G - v$ reaches c : $G - v$ is connected, and $\text{ecc}_{G-v}(c) \leq r - 1$. The reverse inequality is Lemma 2.2. $\square$

Lemma 2.5 (Unique eccentric points, II: the converse). *If v is a non-cut vertex and $r(G - v) = r(G) - 1$, then some central vertex c of G has UEP v .*

Proof. Let c be a centre of $G - v$; then $d_G(c, x) \leq d_{G-v}(c, x) \leq r - 1$ for $x \neq v$, while $\text{ecc}_G(c) \geq r(G) = r$ forces $d_G(c, v) \geq r$. Also v has a neighbour w in $G - v$, giving $d_G(c, v) \leq 1 + d_G(c, w) \leq r$. Hence $d_G(c, v) = r = \text{ecc}_G(c)$: c is central and v is its UEP. $\square$

Lemma 2.6 (A vertex has at most one UEP). *If $x \neq x'$ are both UEPs of c , then $d(c, x') < \text{ecc}(c)$ (from x being the UEP) contradicts $d(c, x') = \text{ecc}(c)$ (from x' being the UEP).*

Lemma 2.7 (Every connected graph has at least two non-cut vertices). *The first and last vertices of a longest path are non-cut.*

Proof. Let $v_1 \dots v_k$ be a longest path and $v = v_1$. Every neighbour of v lies on it, else the path could be extended. For $x \neq v$, follow a geodesic $x = x_0, \dots, x_m = v$; the predecessor x_{m-1} is a neighbour of v , hence equals some v_i with $i \geq 2$. Then $x_0, \dots, x_{m-1}, v_i, v_{i-1}, \dots, v_2$ is a walk in $G - v$ from x to v_2 . So $G - v$ is connected. $\square$

Lemma 2.8. *If $r(G) \geq 2$ then $|V(G)| \geq 4$.*

Proof. $r \geq 2$ means no universal vertex; graphs on at most three vertices all have a universal vertex (on three vertices, both P_3 and K_3 do). $\square$

Lemma 2.9 (Counting; heredity). (i) For $s, \ell \geq 1$: $(s+1)(\ell+1) - 2(s+\ell) = (s-1)(\ell-1) \geq 0$.
(ii) $t(G-v) \leq t(G)$.

Lemma 2.10 (Auxiliary lemma). Let D be a finite simple graph with minimum degree at least one, and let $a \in V(D)$. If every vertex $v \neq a$ has a neighbour of degree one, then a also has a neighbour of degree one.

Proof. Suppose a has no neighbour of degree one, and let $x \in N(a)$; then $\deg(x) \geq 2$. Since $x \neq a$, x has a neighbour y of degree one, so $N(y) = \{x\}$. If $y \neq a$, applying the hypothesis to y forces $\deg(x) = 1$, a contradiction; hence $y = a$ and $\deg(a) = 1$. Pick $z \in N(x) \setminus \{a\}$, and let w be a neighbour of z of degree one ($z \neq a$). Then $w \neq a$ (as $N(a) = \{x\} \not\ni z$) and $w \neq x$ (as $\deg(x) \geq 2$), so $N(w) = \{z\}$; applying the hypothesis to w forces $\deg(z) = 1$, contradicting z being adjacent to the two distinct vertices x and w . $\square$

3 The rooted Chung lemma

Lemma 3.1 (Rooted Chung lemma). Let F be connected with $r(F) = r \geq 2$, and let a be a non-cut vertex with $r(F-a) = r-1$. Then F contains an induced path on at least $2r-1$ vertices that contains a .

Proof. Let v_0 be a centre of $F-a$, so $\text{ecc}_{F-a}(v_0) = r-1$ and $v_0 \neq a$. Every $x \neq a$ satisfies $d_F(v_0, x) \leq d_{F-a}(v_0, x) \leq r-1$.

Claim 0: $d_F(v_0, a) = r$. Since $\text{ecc}_F(v_0) \geq r(F) = r$ while all other vertices are at distance at most $r-1$, we get $d_F(v_0, a) \geq r$. Conversely a has a neighbour $w \neq a$ (as $|V(F)| \geq 2$ and F is connected), and $d_F(v_0, a) \leq 1 + d_F(v_0, w) \leq r$. (This uses that a is non-isolated; the inequality $\leq r$ is the first of two details that are easy to omit.)

Fix a geodesic $v_0, v_1, \dots, v_r = a$; it is induced (Lemma 2.1) and $d(v_0, v_j) = j$.

Since $\text{ecc}_F(v_2) \geq r$, there is a vertex w with $d_F(v_2, w) \geq r$; then $w \neq a$ because $d(v_2, a) = r-2 < r$. Now

$$d_F(v_0, w) \geq d(v_2, w) - d(v_0, v_2) \geq r-2, \quad d_F(v_0, w) \leq r-1,$$

the latter because $w \neq a$. Let $P = p_0, \dots, p_m$ be a v_0 - w geodesic ($p_0 = v_0, p_m = w, m = d(v_0, w) \in \{r-2, r-1\}$); it is induced.

Chord exclusion. (1) For $j \geq 2$, no p_i is adjacent to v_j : otherwise

$$m = d(v_0, w) \geq (j-1) + (d(v_j, w) - 1) \geq (j-1) + (r-j+2-1) = r,$$

using $d(v_j, w) \geq d(v_2, w) - d(v_2, v_j) \geq r - (j-2)$, contradicting $m \leq r-1$. (2) v_1 is not adjacent to any p_i with $i \geq 2$: otherwise, following v_2, v_1, p_i , then P to w , gives $d(v_2, w) \leq 1 + 1 + (m-i) \leq m \leq r-1 < r$, a contradiction. (3) The single possible chord is $v_1 p_1$. Moreover, if $v_1 p_1$ is an edge then $m = r-1$: the same detour as in (2) with $i = 1$ gives $r \leq d(v_2, w) \leq m+1$, i.e. $m \geq r-1$. (This is the second implicit detail: it is what makes the deletion below lose only one vertex.)

Conclusion. Concatenate $Q = v_r, v_{r-1}, \dots, v_0, p_1, \dots, p_m$. By (1) and (2) the only possible chord of Q is $v_1 p_1$, and $a = v_r \in Q$; the order is $(r+1) + m$. If $v_1 p_1$ is not an edge, Q is an induced path of order at least $(r+1) + (r-2) = 2r-1$. If $v_1 p_1$ is an edge, then $m = r-1$, and deleting v_0 from Q leaves $v_r, \dots, v_1, p_1, \dots, p_m$, an induced path of order $r + (r-1) = 2r-1$ (now $v_1 p_1$ is a path edge and, by (2), v_1 has no other chord to P). In both cases the path contains a , which is never deleted. $\square$

Remark 3.2. The classical (unrooted) Erdős–Saks–Sós theorem [1, 2] is *not* used in this paper: the case $\delta \geq 2$ of the conjecture needs only Lemma 2.1, and in the vrd branch the core block is handled with the weaker count of Lemma 2.9(i).

4 The structure theorem for vrd graphs

In this section G is vrd (Definition 1.3) with a cut vertex and $r := r(G) \geq 2$. Write $\text{ncv}(G)$ for the set of non-cut vertices.

Proposition 4.1 (Non-cut vertices are leaves). *Every $v \in \text{ncv}(G)$ is a leaf.*

Proof. Suppose $v \in \text{ncv}(G)$ with $\deg(v) \geq 2$. Let

$$S = \{x : x \text{ is reachable from } v \text{ by a path whose interior and endpoints all lie in } \text{ncv}(G)\},$$

the component of v in $G[\text{ncv}(G)]$. (S contains v and is connected by construction.) S is closed under neighbours. Let $w \in S$.

- If $\deg(w) \geq 2$: by Lemmas 2.5 and 2.4 (with $w \in \text{ncv}$ and $r(G - w) = r - 1$) w is the UEP of a central vertex; then every neighbour x of w satisfies $d(c, x) = r - 1$, and any c - q path through x has length at least $(r - 1) + 1 = r > d(c, q)$, so geodesics from c avoid x ; w has a second neighbour $y \neq x$, which reaches c avoiding x , so $G - x$ is connected: $x \in \text{ncv}(G)$. Being adjacent to $w \in S$, we get $x \in S$.
- If $\deg(w) = 1$: then $w \neq v$ (as $\deg(v) \geq 2$), and the ncv -path from v to w has a last edge into w ; the unique neighbour of w is the penultimate vertex of that path, which lies in S .

So no edge leaves S . But S is nonempty ($v \in S$), proper (it contains only non-cut vertices, while G has a cut vertex), connected, and G is connected — a contradiction. $\square$

Corollary 4.2. (i) G has a leaf (indeed $\text{ncv}(G)$ consists of leaves, and $|\text{ncv}| \geq 2$ by Lemma 2.7). (ii) $r \geq 2$ (assumed). $\square$

Proposition 4.3 (Leaf stripping). *Let L be the set of leaves of G and $G_1 = G - L$. Then G_1 is connected and $r(G_1) = r - 1$. Moreover every vertex of G_1 has degree at least two in G_1 .*

Proof. G_1 is connected (deleted vertices are leaves). Lower bound: a centre c of G_1 satisfies $d_G(c, x) = d_{G_1}(c, x) \leq r(G_1)$ for non-leaves x (paths avoid leaves, Lemma 2.3(i)) and $d_G(c, z) \leq 1 + r(G_1)$ for leaves z ; hence $r(G) \leq r(G_1) + 1$, i.e. $r(G_1) \geq r - 1$. Upper bound: $r(G - u) = r - 1$ for a leaf u (Definition 1.3 and Lemma 2.2), and deleting the remaining leaves one by one never increases the radius (Lemma 2.3(ii)). $\square$

Proposition 4.4 (The penultimate layer). *Let L, G_1 be as above and let w be a non-cut vertex of G_1 . Then w is a cut vertex of G and has at least one leaf neighbour.*

Proof. $w \notin L$ (leaves of G are isolated in G_1 , and $|V(G_1)| \geq 2$). If w were a non-cut vertex of G , Proposition 4.1 would make w a leaf, a contradiction; so w is a cut vertex of G . Now $G - w$ is disconnected, while $G_1 - w$ is connected; so $G_1 - w$ lies inside one component of $G - w$, and every other component consists of leaves only. A component of $G - w$ consisting of leaves is a single vertex adjacent to w (two leaves cannot be adjacent: each would have its neighbour outside the component). Such components exist because $G - w$ is disconnected, so w has a leaf neighbour. $\square$

Proposition 4.5 (Two leaves do not share a parent). *A vertex of G has at most one leaf neighbour.*

Proof. Suppose leaves u, u' share the parent b . By Lemma 2.5 (applied to the non-cut vertex u) some central c has UEP u , so $d(c, b) = r - 1$ and $c \neq b$ (else $r = d(b, u) = 1$). Every c - u' path ends with $b u'$, so $d(c, u') = d(c, b) + 1 = r$, contradicting the uniqueness in UEP(c, u) (as $u' \neq u$). $\square$

Proposition 4.6 (Stripping preserves vrd). *$G_1 = G - L$ is vrd: for every non-cut vertex w of G_1 , $r(G_1 - w) = r(G_1) - 1 = r - 2$.*

Proof. Fix such w . By Proposition 4.4, w has a leaf neighbour u ; by Proposition 4.5 it is the unique one. Let c be central in G with UEP(c, u) (Lemmas 2.5 applied to $u \in \text{ncv}(G)$): $d(c, u) = r$, all others $\leq r - 1$, so $d(c, w) = r - 1$; also $c \notin L$ (Lemma 2.3(iii)), so $c \in V(G_1)$ and $\text{ecc}_{G_1}(c) = r - 1 = r(G_1)$.

Claim: $d(c, z) \leq r - 2$ for all $z \in V(G_1) - w$. Suppose $d(c, z) = r - 1$ (distances agree with G by Lemma 2.3(i)). Then $z \notin L$ and $z \neq w$, so z is a cut vertex of G (otherwise Proposition 4.1 puts z in L). Let $C_0 \ni c$ be the component of $G - z$ and let C be another component, nonempty. For $y \in C$, every c - y path passes through z , so $d(c, y) \geq d(c, z) + 1 = r$; since $\text{ecc}(c) = r$, we get $d(c, y) = r$, hence $y = u$ by UEP(c, u). So $C = \{u\}$. But w is adjacent to u and $w \neq z$, so $w \in C$, i.e. $w = u$ — absurd (as u is a leaf and w has degree at least two in G_1). This proves the claim.

Hence $\text{ecc}_{G_1-w}(c) \leq r - 2$, so $r(G_1 - w) \leq r - 2$; the reverse bound is Lemma 2.2 applied in G_1 . $\square$

Proposition 4.7 (Decreasing blocks). *Let B be a connected graph with no cut vertex such that $r(B - v) = r(B) - 1$ for every v (e.g. the terminal graph of the stripping chain, by Proposition 4.6). Then B is self-centered and admits a fixed-point-free involution $f : V(B) \rightarrow V(B)$ with UEP($f(v), v$) for every v ; in particular each vertex has a unique eccentric vertex.*

Proof. For each v , Lemma 2.5 gives a central c_v with UEP(c_v, v); by Lemma 2.6 the map $f : v \mapsto c_v$ is injective. Hence $|V| \leq |C(B)| \leq |V|$, so $C(B) = V(B)$: B is self-centered, and f is a bijection with $f(v) \neq v$ (as $d(v, f(v)) = r(B) \geq 1$). If $f(f(v)) \neq v$, then $f(f(v))$ and v are both at distance $r(B)$ from $f(v)$ (the former because UEP($f^2(v), f(v)$) says $d(f^2(v), f(v)) = r(B)$), contradicting uniqueness of the eccentric vertex of $f(v)$; so $f^2 = \text{id}$. Finally UEP($v, f(v)$) is the statement UEP($f(f(v)), f(v)$) read at v . $\square$

Theorem 4.8 (Structure, self-contained version of Theorem 1.5). *$G = S_k(B)$ for the terminal block B of the stripping chain and $k = r - r(B) \geq 1$; in particular $r(G) = r(B) + k$.*

Proof. Form the stripping chain $G = G_0 \supset G_1 \supset \dots$, $G_{i+1} = G_i - L(G_i)$ ($L(\cdot)$ the leaf set). By Propositions 4.3 and 4.6, every G_i with a cut vertex is vrd and $r(G_i) = r - i$; since $|V(G_i)|$ strictly decreases, the chain terminates at the first k with $G_k =: B$ having no cut vertex, and $r(B) = r - k \geq 1$, so $|V(B)| \geq 2$.

We show by downward induction on i that $G_i = S_{k-i}(B)$. For $i = k$ this holds with $S_0(B) := B$. Assume $G_{i+1} = S_m(B)$ with $m = k - i - 1 \geq 0$, and let P be the set of parents of $L(G_i)$; each leaf of G_i has its parent in G_{i+1} (a leaf whose parent were also deleted would force $G_i = K_2$, whose radius 1 contradicts $r(G_i) = r - i \geq r - k + 1 \geq 2$), and each vertex has at most one leaf (Proposition 4.5). We must show that P is exactly the set of the $|B|$ tips of $G_{i+1} = S_m(B)$ (the vertices at depth m , one on each pendant path).

(a) *Every tip is in P .* Since $r(G_i) = r(B) + m + 1$, every $z \in B$ satisfies $\text{ecc}_{G_i}(z) \geq r(B) + m + 1$. Within $G_{i+1} = S_m(B)$, a vertex at depth j on the branch over b' is at distance at most $d(z, b') + j \leq r(B) + m$ from z (as $d(z, b') \leq r(B)$, B being self-centered). So some $p \in P$ has $d(z, p) \geq r(B) + m$;

also $d(z, p) \leq r(B) + \text{depth}(p) \leq r(B) + m$. Hence $d(z, p) = r(B) + m$ and the witness is unique: B has a unique eccentric vertex $f(z)$ (Proposition 4.7), and the distance bound shows p lies on $f(z)$'s branch at depth m , i.e. p is $f(z)$'s tip. As f is surjective, every tip is in P .

(b) *No other vertex is in P .* Suppose $p \in P$ has depth $j < m$ on the branch over $b \in B$, with leaf u adjacent to p . The vertex u is a non-cut vertex of G_i , so by Lemma 2.5 some centre c of G_i has $\text{UEP}(c, u)$: $d(c, u) = r(B) + m + 1$, hence $d(c, p) = r(B) + m$. Now c is not a leaf of G_i (Lemma 2.3(iii)). Locate c :

- $c \in B$: then $d(c, p) \leq r(B) + j \leq r(B) + m - 1$, too small.
- c on p 's own branch at depth j' : then $d(c, p) = |j' - j| \leq m$, too small.
- c on the branch over $b'' \neq b$ at depth j'' : let w be the tip of p 's branch (depth m ; it exists and lies in G_{i+1}). Then

$$d(c, w) = j'' + d(b'', b) + m = d(c, p) + (m - j) \geq r(B) + m + 1 = \text{ecc}(c),$$

so $d(c, w) = \text{ecc}(c)$, and uniqueness in $\text{UEP}(c, u)$ forces $w = u$; but $w \in V(G_{i+1})$ while $u \notin V(G_{i+1})$, a contradiction.

This completes the induction: $G = G_0 = S_k(B)$, and $r(G) = r(B) + k$ because the stripping chain drops the radius by exactly one per step (Propositions 4.3 and 4.6). Moreover every vertex of B has degree $\ell + 1$ in G for the appropriate layer, as asserted in Theorem 1.5. $\square$

Corollary 4.9 ($t \geq 2r$ for vrd graphs with a cut vertex). *If G is vrd with a cut vertex, then $t(G) \geq 2r(G)$.*

Proof. Write $G = S_k(B)$ as in Theorem 4.8, $s = r(B) \geq 1$. Take a geodesic Q of B (order at least $s + 1$, Lemma 2.1). The union of $V(Q)$ with, for each $q \in Q$, the ℓ -vertex pendant path attached to q , induces a tree: pendant paths are internally disjoint, there are no edges between distinct pendant paths, and Q is induced in B (Lemma 2.1); its order is at least

$$(s + 1)(k + 1) \geq 2(s + k) = 2r(G)$$

by Lemma 2.9(i). $\square$

Remark 4.10. The case $B = K_2$ needs no special treatment: K_2 satisfies Proposition 4.7 with f the transposition, and equality holds throughout Corollary 4.9 ($G = P_{2k+2}$, $t = 2r$).

Remark 4.11 (Historical credit). Theorems of this shape are classical: Glivak [4] (see [5], Theorem 4, and Swart [6, Theorem 2.2.10]) describes radially critical graphs as a “decreasing” core with equilength endpaths, and the fact that non-cut vertices of vrd graphs with a cut vertex are leaves is attributed to Fajtlowicz [3]. The proofs above were reconstructed from first principles and are self-contained; we did not consult the original arguments.

5 The leaf lemma

Proof of Lemma 1.2. Suppose the lemma fails and let G be a counterexample of minimum order: G connected with a leaf u and $t(G) < 2r(G)$, while every smaller connected graph with a leaf satisfies the lemma. Let $a = p(u)$ and $r = r(G)$.

If $r = 1$, the edge ua induces a tree on $2 = 2r$ vertices, a contradiction. So $r \geq 2$; by Lemma 2.8 $|V(G)| \geq 4$, and a is a cut vertex (Lemma 2.3(iv)).

Claim 1. For every non-cut vertex $v \neq u$ of G : $r(G - v) = r - 1$. Indeed $v \neq a$ (as a is a cut vertex and v is not), so $G - v$ is connected and still has the leaf u . If $r(G - v) \geq r$, minimality gives $t(G - v) \geq 2r(G - v) \geq 2r$, while $t(G - v) \leq t(G)$ (Lemma 2.9(ii)) — a contradiction. So $r(G - v) < r$, and Lemma 2.2 gives $r(G - v) = r - 1$.

Setup. Let $H = G - u$. By Lemma 2.3, distances among $V(H)$ agree in G and H , and u is not central; hence centres of G survive in H and $r(H) \leq r$. If $r(H) < r$, then u is a non-cut vertex and Lemma 2.2 gives $r(H) = r - 1$. So $r(H) \in \{r - 1, r\}$.

Case A: $r(H) = r - 1$. Then every non-cut vertex of G deletes the radius to $r - 1$: u by the case assumption, all others by Claim 1. So G is vrd with the cut vertex a , and Corollary 4.9 gives $t(G) \geq 2r$ — a contradiction. (Minimality is not needed in this case.)

Case B: $r(H) = r$.

Claim 2. H has no leaf; in particular $\delta(H) \geq 2$. Otherwise H is a smaller connected graph with a leaf and $r(H) = r$, so minimality gives $t(H) \geq 2r(H) = 2r$; but $t(G) \geq t(H)$ (Lemma 2.9(ii)), contradiction.

Claim 3. For every non-cut vertex $v \neq a$ of H : $r(H - v) = r - 1$. $G - v$ is obtained from the connected graph $H - v$ by re-attaching the leaf u at a , hence is connected; so v is a non-cut vertex of G , and Claim 1 gives $r(G - v) = r - 1$. The graph $G - v$ has at least three vertices, so its leaf u is not central (Lemma 2.3(iii)); hence centres of $G - v$ survive in $H - v$ and $r(H - v) \leq r - 1$. The reverse inequality is Lemma 2.2 (as $r(H) = r$ and v is a non-cut vertex of H).

Claim 4. H has no cut vertex. Claims 2 and 3 and Lemma 2.5 imply that every non-cut vertex $v \neq a$ is a central vertex's UEP and has degree at least two. The neighbour argument in Proposition 4.1 therefore shows that every neighbour of v is also a non-cut vertex. Choose a non-cut vertex $v \neq a$; two distinct non-cut vertices exist by the longest-path argument. If a were a cut vertex, the non-cut property would propagate along any path starting at v : a non-cut vertex can never equal a , so the neighbour implication applies at every step. A path from v to a would give a contradiction. Thus a is a non-cut vertex. For any $w \neq a$, choose a v - w path in the connected graph $H - a$. Propagating the same implication along this path shows that w is a non-cut vertex. Hence every vertex of H is a non-cut vertex.

By Claim 4 every vertex of H is a non-cut vertex. For each $v \neq a$, Claim 3 and Lemma 2.5 provide a central c_v with UEP(c_v, v); distinct v give distinct c_v (Lemma 2.6). So with $n = |V(H)|$, at least $n - 1$ vertices of H are central.

Claim 5. H is self-centered. If some b were non-central, then every $y \neq b$ is central, whence $d(y, b) \leq r$, so $d(b, y) \leq r$ and $\text{ecc}(b) \leq r$; but always $\text{ecc}(b) \geq r(H) = r$. Hence $\text{ecc}(b) = r$ and b is central, a contradiction.

Form the auxiliary graph D on $V(H)$ by putting $xy \in E(D)$ iff $d_H(x, y) = r$. Since H is self-centered and finite, every vertex has a vertex at distance exactly r : $\delta(D) \geq 1$. For each $v \neq a$, c_v is adjacent to v in D and $\deg_D(c_v) = 1$ (any other w with $d(c_v, w) = r$ would violate UEP(c_v, v)). So every $v \neq a$ has a degree-one neighbour, and Lemma 2.10 provides c adjacent to a in D with $\deg_D(c) = 1$. Thus $d_H(c, a) = r$ and a is the unique vertex at distance r from c ; by Claim 5, c is central, so UEP(c, a). Lemma 2.4 gives $r(H - a) = r - 1$, and a is a non-cut vertex of H (Claim 4). Lemma 3.1 yields an induced path P of H with $|P| \geq 2r - 1$ and $a \in P$.

Finally, u is adjacent in G only to $a \in P$, so $G[V(P) \cup \{u\}]$ is an induced tree on at least $2r$ vertices — the final contradiction. $\square$

6 Graffiti.pc Conjecture 84

Theorem 6.1. *Let G be a finite connected graph with at least two vertices. Then*

$$2r(G) \leq t(G)\delta(G),$$

equivalently $t(G) \geq 2r(G)/\delta(G)$.

Proof. A connected nontrivial graph has $r(G) \geq 1$ and $\delta(G) \geq 1$. If $\delta(G) \geq 2$, Lemma 2.1 gives $t(G) \geq r(G) + 1$, so $t(G)\delta(G) \geq 2(r(G) + 1) > 2r(G)$. If $\delta(G) = 1$, G has a leaf, and Lemma 1.2 gives $t(G) \geq 2r(G)$, so $t(G)\delta(G) = t(G) \geq 2r(G)$. $\square$

Remark 6.2. The leaf lemma is genuinely about induced trees: one cannot strengthen it to an induced *path* on $2r$ vertices. In Case B of its proof the extra vertex is a pendant attachment to a long induced path; in Case A the tree is assembled from several pendant branches over a geodesic of the core block.

7 Formalization notes

The proof decomposes into Lean modules as follows.

- (i) elementary facts (Section 2): geodesics, radius bookkeeping, UEP drop and converse, leaves;
- (ii) the rooted Chung lemma (Lemma 3.1);
- (iii) the structure theorem (Section 4): Propositions 4.1–4.7 and Theorem 4.8;
- (iv) the leaf lemma (Section 5);
- (v) the arithmetic assembly (Theorem 6.1).

The Case B non-cut argument now uses only paths and the existence of two non-cut vertices, eliminating the former block-tree prerequisite. The companion Lean package follows a shorter route for Case A: direct induction by peeling leaves, without the full structural classification presented in this manuscript. Its final declarations are `Graffiti84.leafLemma` and `Graffiti84.graphConjecture84`. The exact commit's GitHub Actions run checks the declarations and prints their axiom dependencies. A guide to the formal proof is kept in `docs/formal-proof-guide.md`; this manuscript retains the historical classification route.

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